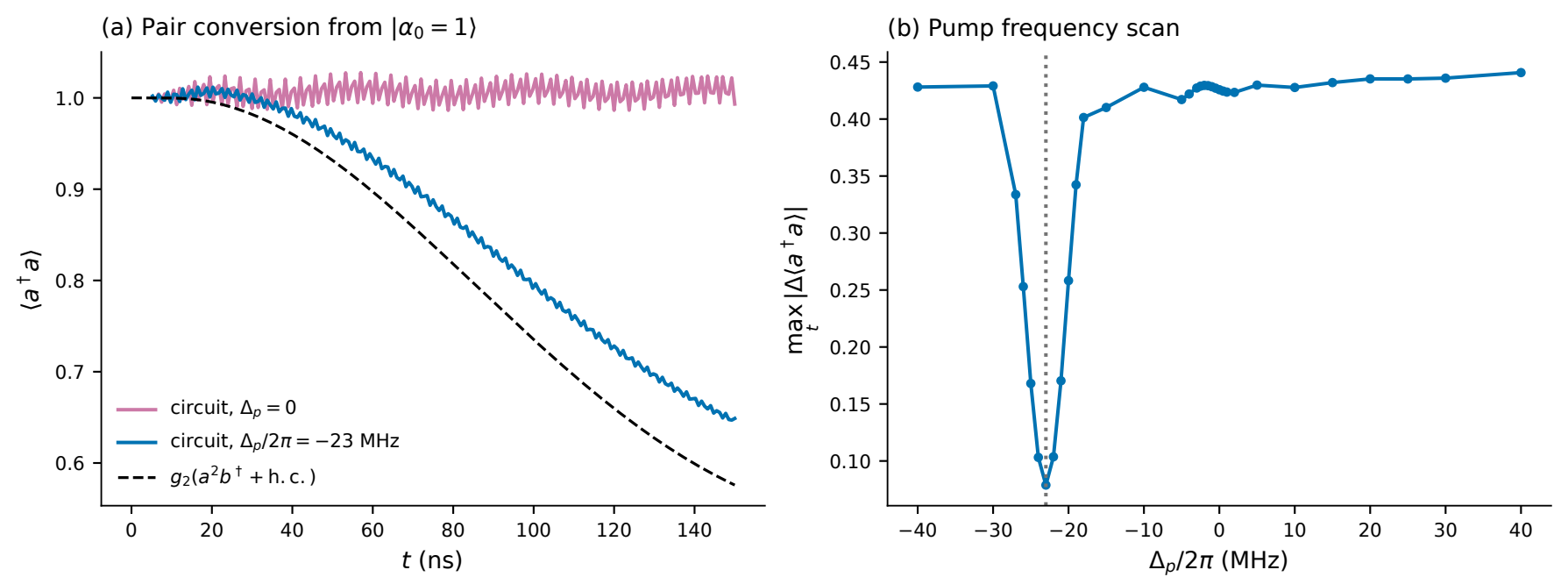
(a) Pair conversion from $|\alpha_0 = 1\rangle$
$\langle a^\dagger a \rangle$
1.0
0.9
0.8
0.7
0.6
0
20
40
60
80
100
120
140
t (ns)
circuit, $\Delta_p = 0$
circuit, $\Delta_p/2\pi = -23$ MHz
$g_2(a^2 b^\dagger + \text{h. c.})$
(b) Pump frequency scan
$\max_t |\Delta\langle a^\dagger a \rangle|$
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
−40
−30
−20
−10
0
10
20
30
40
$\Delta_p/2\pi$ (MHz)